

Lecture 6: Architectures and Practical Training



Today:

00:00) First architectures (LeNet, AlexNet)

00:20) Initialization & activation functions

00:45) ---- Break ----

00:55) Optimizers: SGD, Momentum, Adam, Muon

01:20) Regularization: weight decay, dropout, Mixup

01:40) ---- Break ----

01:50) Depth, ResNet & residual connections

02:05) Normalization: BatchNorm, LayerNorm, RMSNorm

02:30) U-Net, stability tricks & ConvNeXt

02:50) Attention & Vision Transformers

03:10) Debugging training: a practical playbook

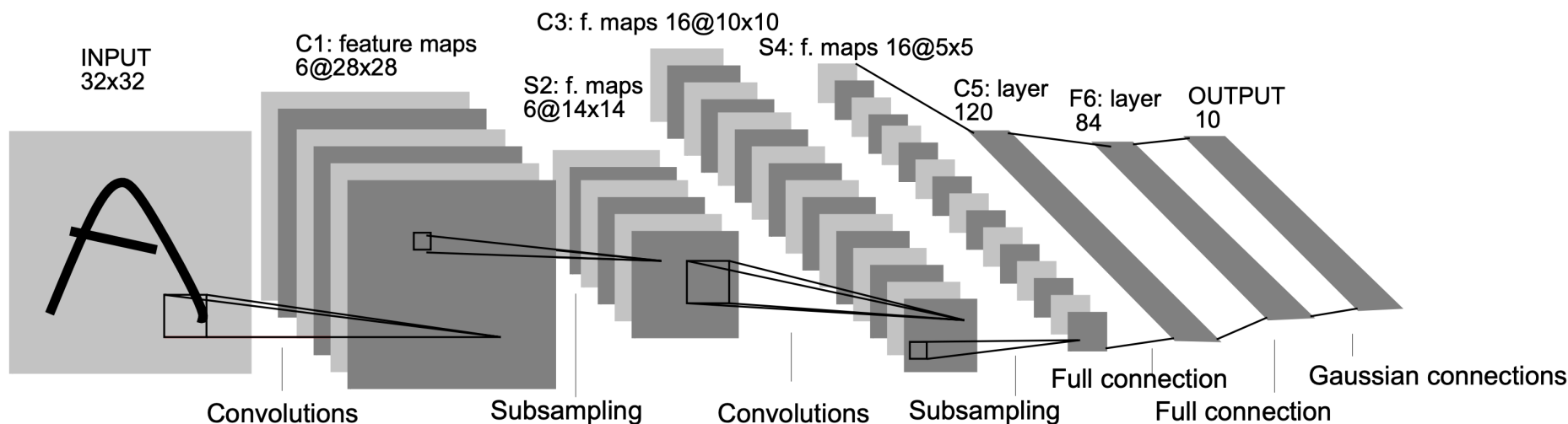
03:30) Tutorial: Transformers & ViTs

LeNet 5 1998



Yann LeCun

- conv → pool → fully-connected.
- All the ingredients were already here in '98. (or even '89. or even before?)



TOOLBOX

what we can build so far

1 / 22 acquired

ARCHITECTURE

- conv · pool · FC
- residual block
- U-Net skips
- attention / ViT

ACTIVATIONS

- ReLU
- GELU / SiLU

NORMALIZATION

- BatchNorm
- Layer / Group
- RMSNorm · DyT

REGULARIZATION

- L2
- dropout
- dropPath
- Weight decay
- augmentation
- label smooth · Mixup

OPTIMIZATION

- SGD + momentum
- Nesterov
- Adam / AdamW
- Muon

GRADIENT TRICKS

- straight-through est.
- gradient clipping
- weight EMA

Initialization

Start with all zeros? 🙅

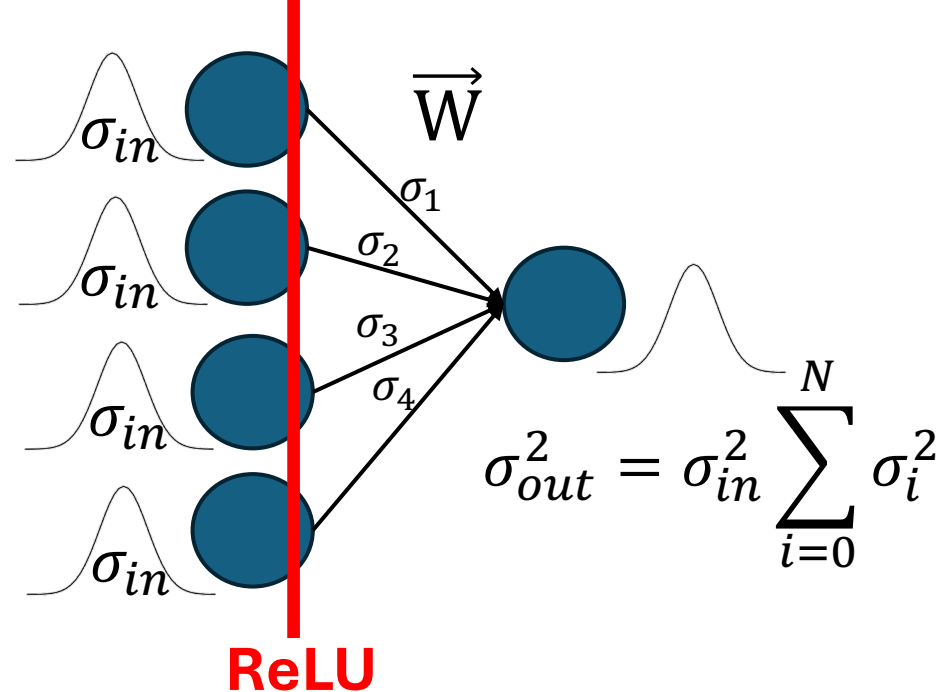
Goal: Variance preservation

$$\sigma_{out}^2 = \sigma_{in}^2$$

$$\sum_{i=0}^N \sigma_i^2 = 1$$

$$\sigma_i = \frac{1}{\sqrt{N}}$$

Xavier init



$$\sigma_{relu}^2 = \frac{1}{2} \sigma_{in}^2$$

$$\sigma_i = \frac{1}{\sqrt{2N}}$$

Kaiming init
(yes, him again)

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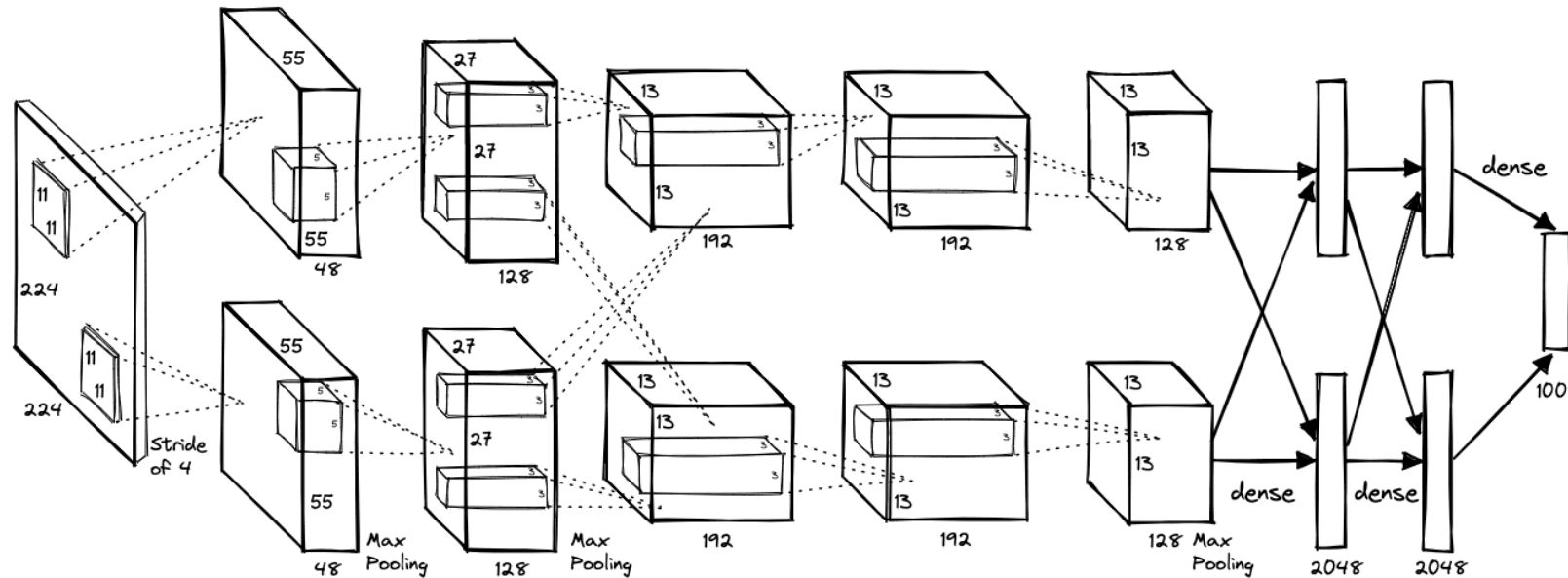
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AlexNet: it works at scale 2012



Ilya Sutskever Geoffrey Hinton Alex Krizhevsky

- ReLU
- Data augmentation
- GPUs



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Activations: Vanishing grads and Dead Neurons

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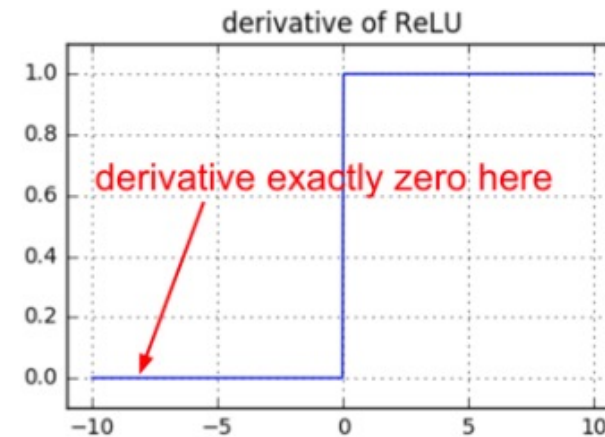
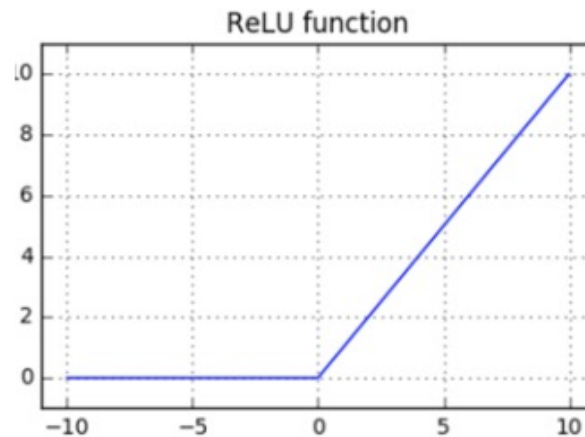
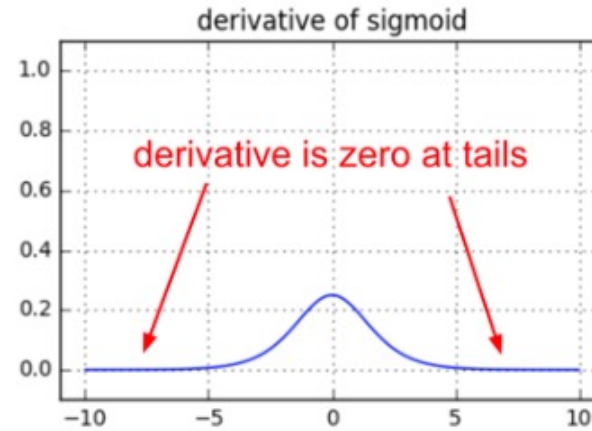
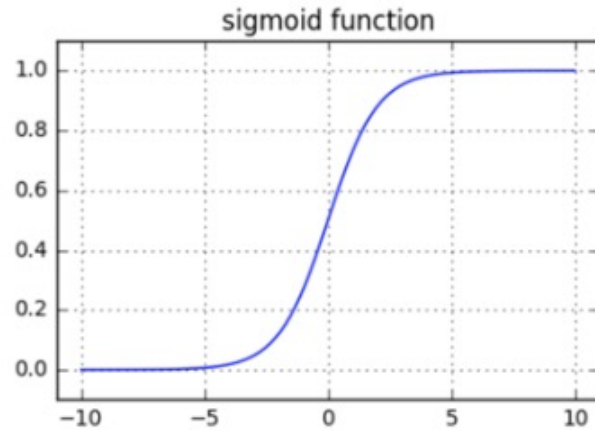
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Activation functions

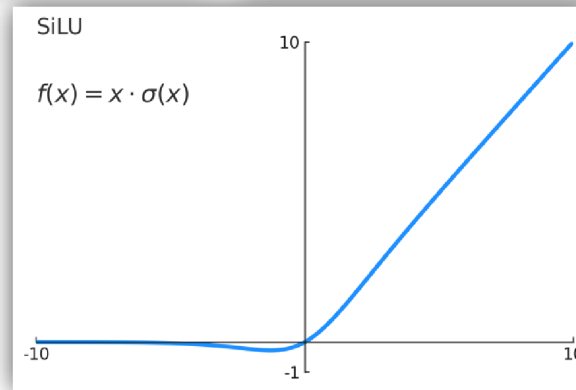
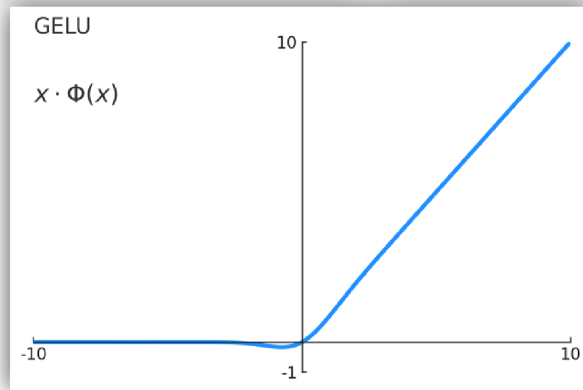
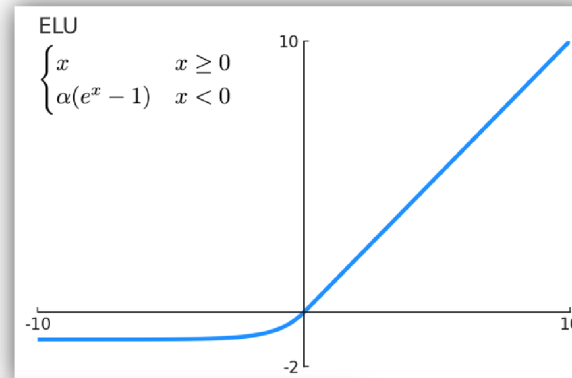
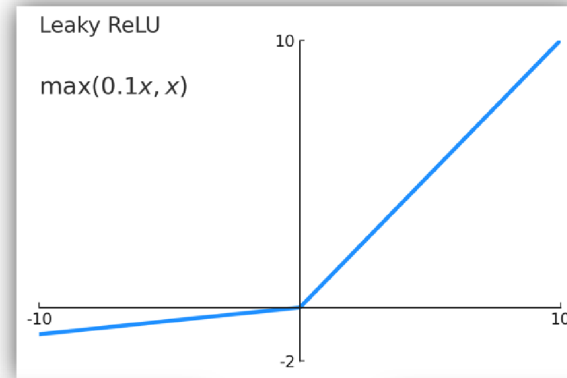
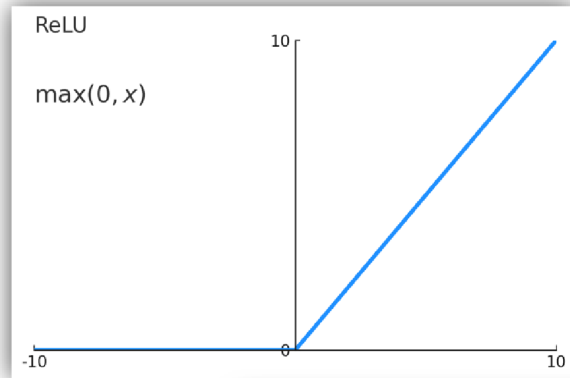


Figure credit: CS231n; Fei-Fei & Adeli & Durante & Xiang & Johnson

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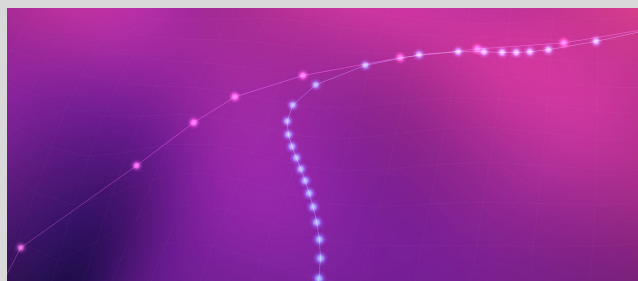
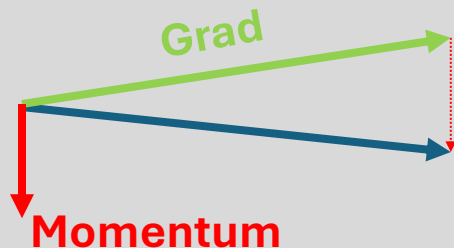
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Optimizers

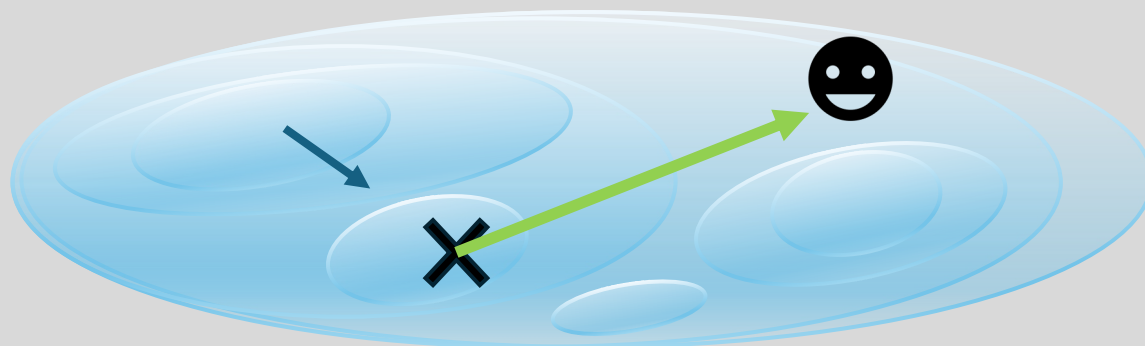
Vanilla SGD: $w_i := w_i - \alpha \frac{d\mathcal{L}(w)}{dw_i}$

SGD + Momentum: $m_i := \gamma m_i + \alpha \frac{d\mathcal{L}(w)}{dw_i}$
 $w_i := w_i - m_i$

Typical: $\gamma = 0.9$
**Exponential
Moving
Average**



Nesterov Accelerated Gradients : $m_i := \gamma m_i + \alpha \frac{d\mathcal{L}(w + \gamma m)}{dw_i}$



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6 / 22 acquired

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Optimizers

Adaptive Moment Est. (ADAM):

$$m_i := \beta_1 m_i + (1 - \beta_1) \frac{d\mathcal{L}}{dw_i}$$

$$v_i := \beta_2 m_i + (1 - \beta_2) \left(\frac{d\mathcal{L}}{dw_i} \right)^2$$

$$w_i := w_i - \alpha \frac{\hat{m}}{\sqrt{\hat{v}} + \varepsilon}$$

MomentUm Orthogonalized by Newton-schulz (Muon):

$$O = \text{NS}(M) \approx UV^\top$$

$$w_i := w_i - \alpha O$$

the gradient is a **matrix** so treat it like one.

Newton-Schulz = matmuls, no SVD.

$$X \leftarrow aX + b(XX^\top)X + c(XX^\top)^2X$$

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8 / 22 acquired

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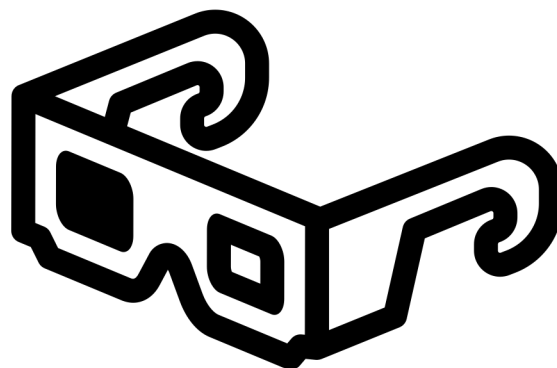
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Demo:



Regularization

Regularization loss term:

$$\text{E.g, } \mathcal{L}' = \mathcal{L} + \lambda \sum w_i^2$$

- L2 regularization is communist!



Weight decay: at each update do

$$w := w(1 - \beta). \quad (\text{small } \alpha)$$

Actually, they are (almost) the same!

$$\begin{aligned} w_i &:= w_i - \alpha \frac{d\mathcal{L}'}{dw_i} \\ &= w_i - \alpha \left(\frac{d\mathcal{L}}{dw_i} + \lambda \sum \frac{dw_j^2}{dw_i} \right) \\ &= \boxed{(1 - 2\lambda)w_i - \alpha \frac{d\mathcal{L}}{dw_i}} \end{aligned}$$

$\mathcal{L}^{1/2}$

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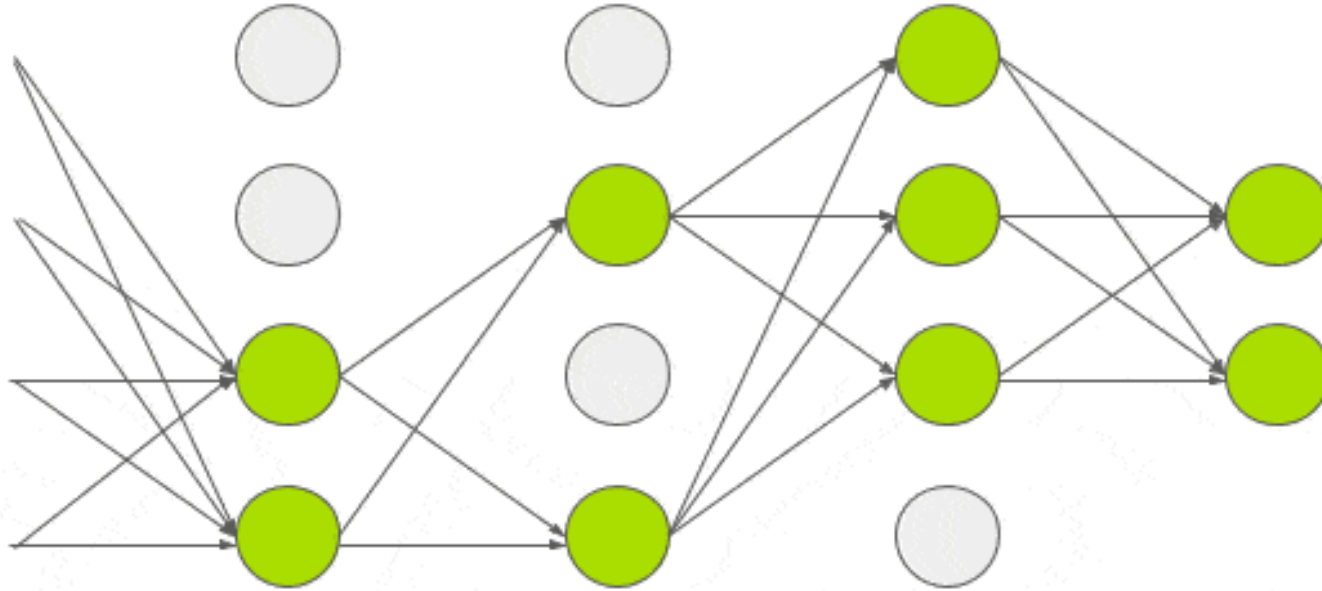
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Dropout



- Regularization.
- Can't “cheat” using too little information.
- Trains to be robust to differences, missing info.
- At inference, all neurons are on. Scaling needed (in PT built-in)

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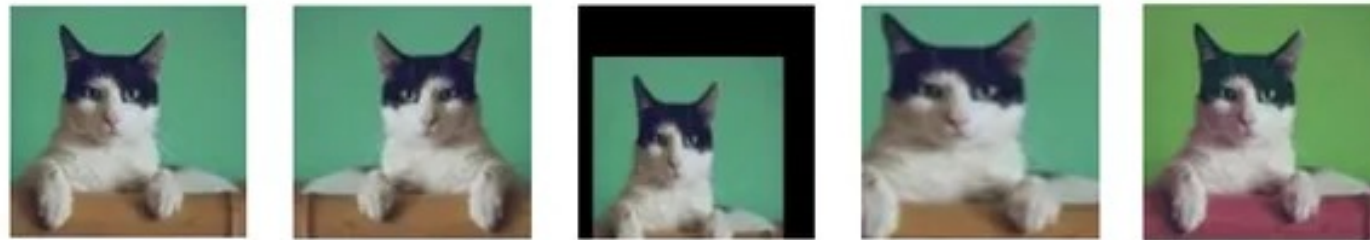
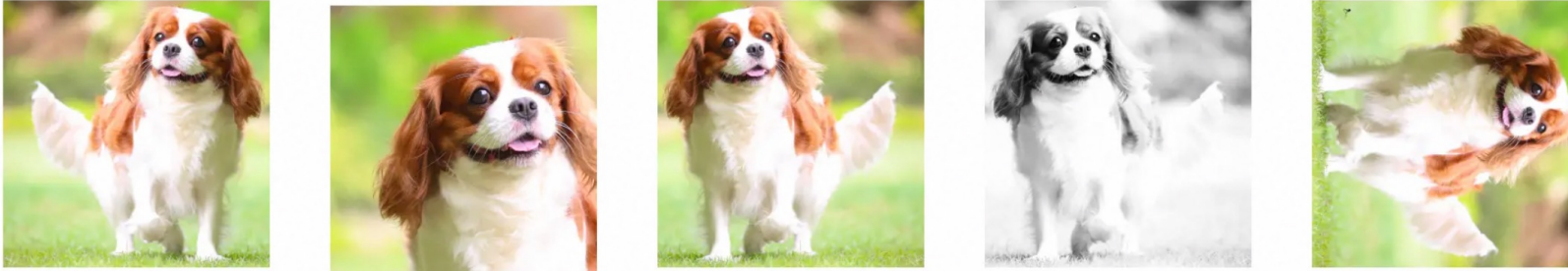
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Data augmentation



Input

Flip

Crop

Zoom

Color change



Cutout



Mixup



Cutmix



Mosaic



Dynamic Mosaic

Soft
labels

Object
detection

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12 / 22 acquired

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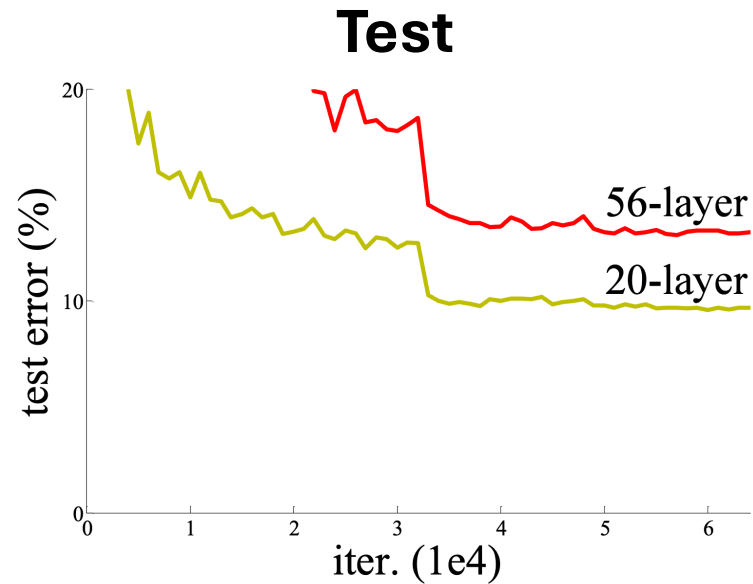
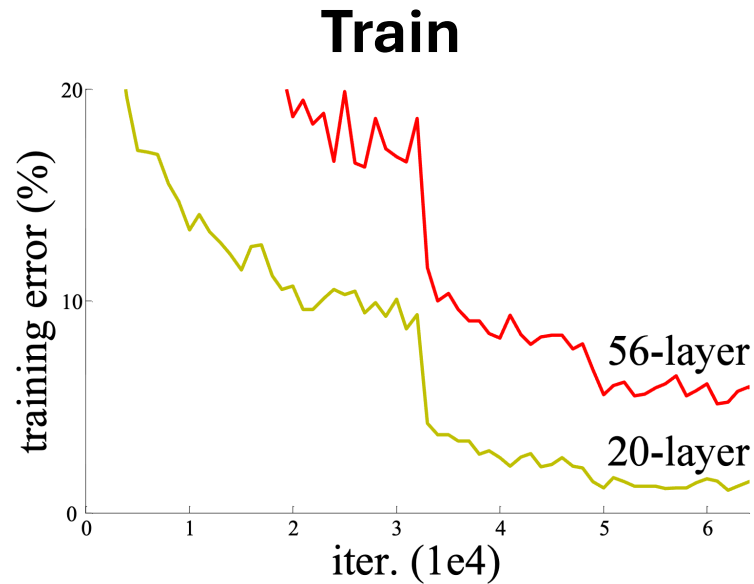
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The effect of depth



This is not overfitting! Then what is this?

Optimization: vanishing gradients

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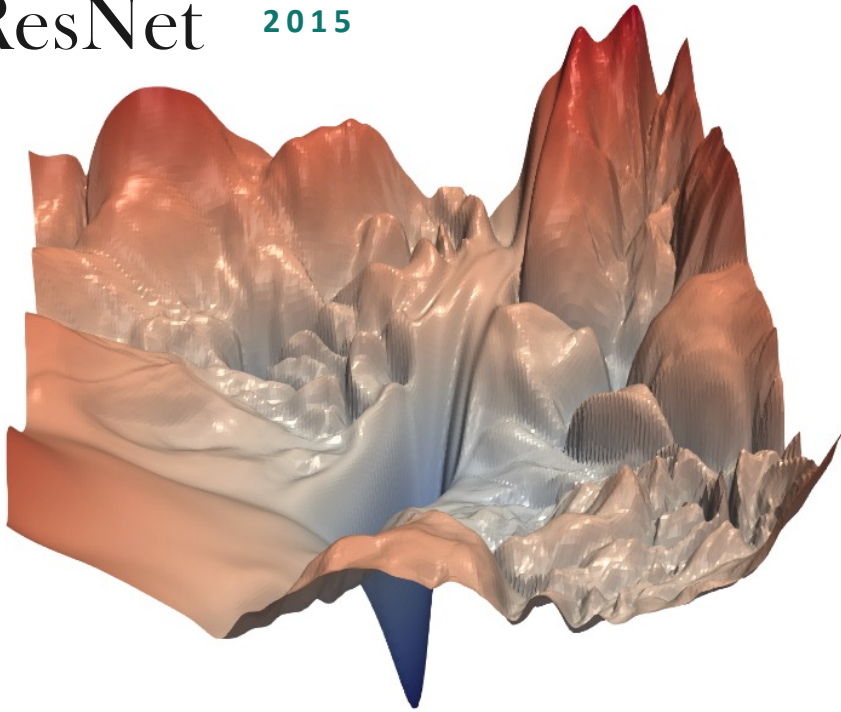
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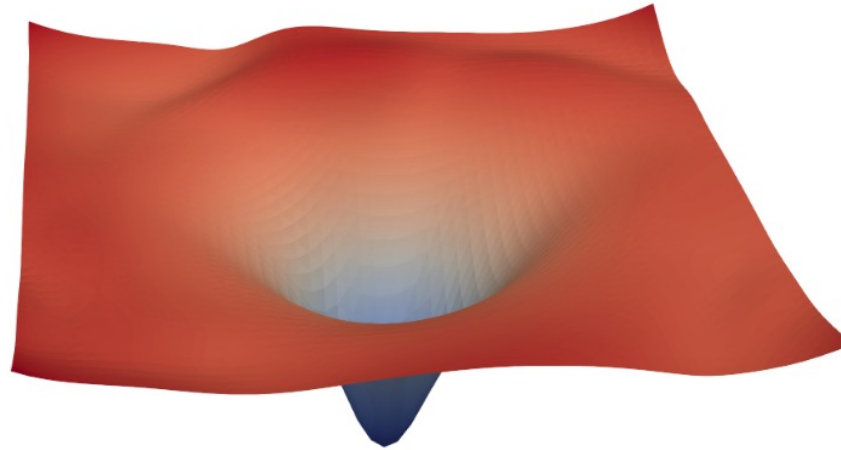
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ResNet 2015



(a) without skip connections



(b) with skip connections



Kaiming He

Associate Professor, EECS, [MIT](#)
Verified email at mit.edu - [Homepage](#)
[Computer Vision](#) [Machine Learning](#)



TITLE

[Deep Residual Learning for Image Recognition](#)

K He, X Zhang, S Ren, J Sun
Computer Vision and Pattern Recognition (CVPR), 2016

CITED BY

327858

YEAR

2016

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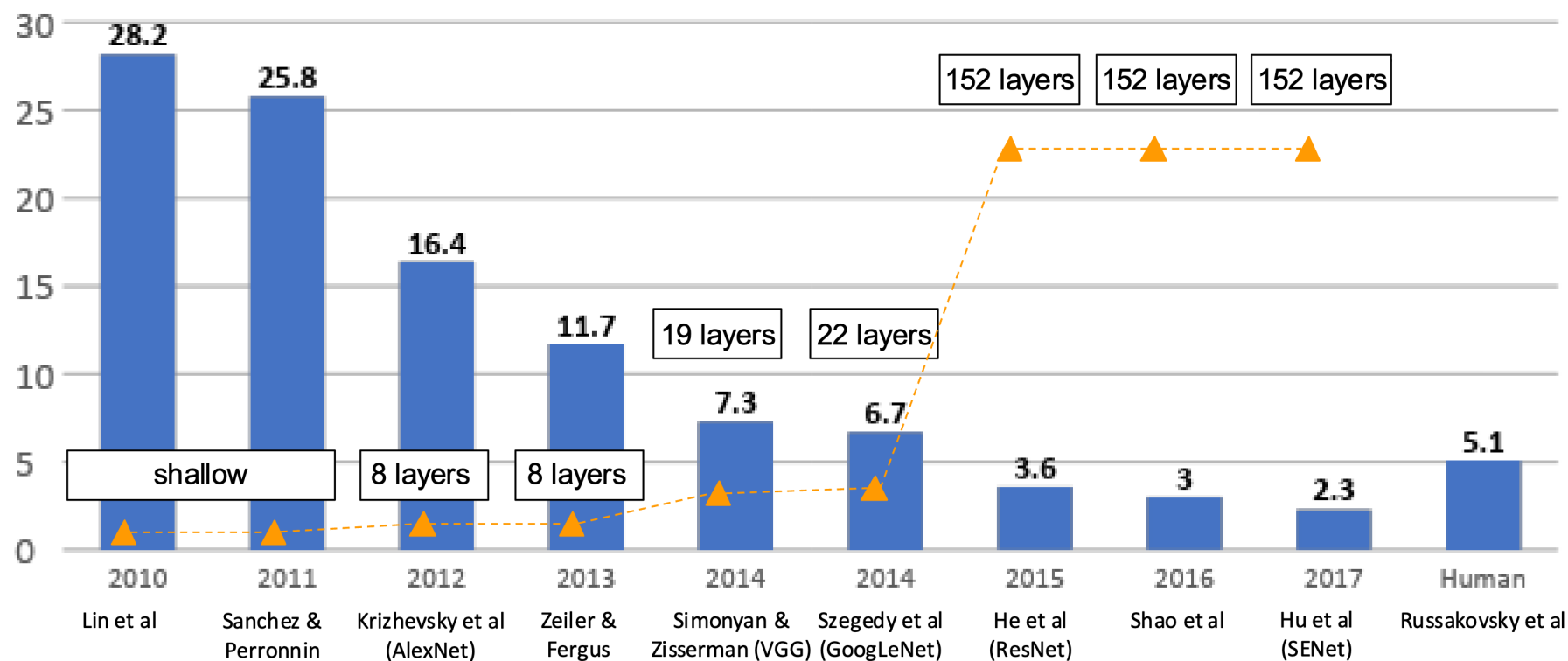
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ImageNet Large Scale Visual Recognition Challenge (ILSVRC) winners



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Normalization

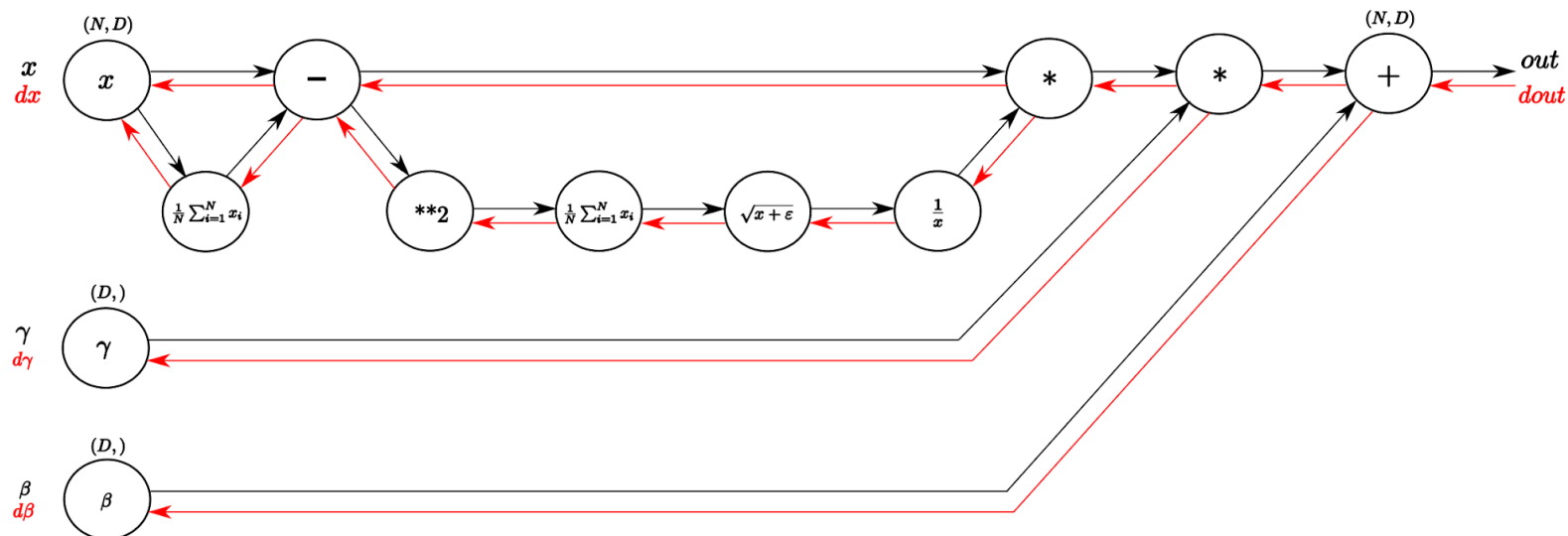
- Common ML practice: normalize input data.
- In DL, every layer is an input to the rest of the net.

Example: BatchNorm

Input: Values of x over a mini-batch: $\mathcal{B} = \{x_{1\dots m}\}$;
Parameters to be learned: γ, β

Output: $\{y_i = \text{BN}_{\gamma, \beta}(x_i)\}$

$$\mu_{\mathcal{B}} \leftarrow \frac{1}{m} \sum_{i=1}^m x_i \quad // \text{ mini-batch mean}$$
$$\sigma_{\mathcal{B}}^2 \leftarrow \frac{1}{m} \sum_{i=1}^m (x_i - \mu_{\mathcal{B}})^2 \quad // \text{ mini-batch variance}$$
$$\hat{x}_i \leftarrow \frac{x_i - \mu_{\mathcal{B}}}{\sqrt{\sigma_{\mathcal{B}}^2 + \epsilon}} \quad // \text{ normalize}$$
$$y_i \leftarrow \gamma \hat{x}_i + \beta \equiv \text{BN}_{\gamma, \beta}(x_i) \quad // \text{ scale and shift}$$



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BatchNorm nuances

Train vs Test:

What happens if at test time we input batch=1?

Solution:

Mean and Var frozen at test.
Calculated via EMA during training

• Training time:

- $\hat{x} = \frac{x - \mu_B}{\sigma_B}$
- $\mu_{EMA} \leftarrow \lambda \mu_{EMA} + (1 - \lambda) \mu_B$
- $\sigma_{EMA}^2 \leftarrow \dots$

• Test time:

- No concept of “batch”
- $\hat{x} = \frac{x - \mu_{EMA}}{\sigma_{EMA}}$

2D vs 1D:

In 2D, normalize over all pixel positions and batch dim
(all dims except channels).

Rational: CNN spirit- batch of patches.

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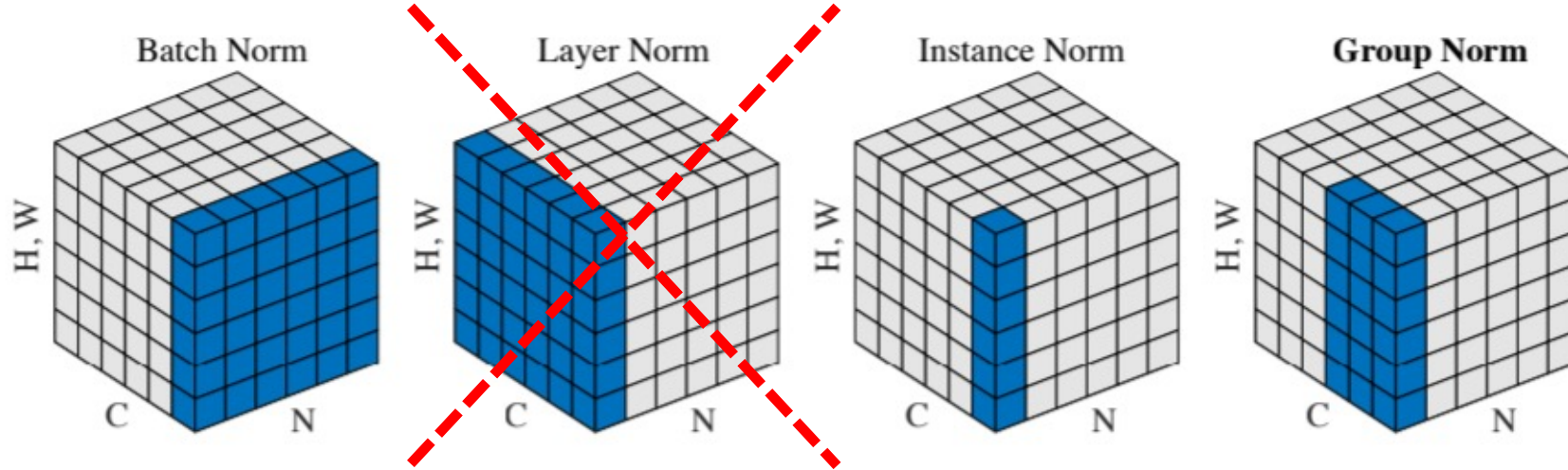
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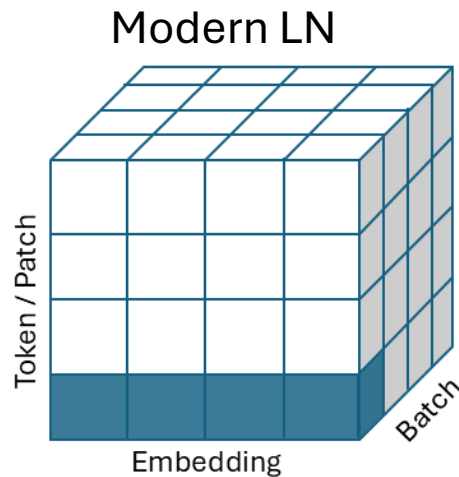
GRADIENT TRICKS

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Other norm layers



LN nuances



In PyTorch:

LayerNorm($d_1 \dots d_N$) reduces last N dims.
In practice when people say LN they mean.
LayerNorm(C): over the channels/embedding.

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15 / 22 acquired

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Dynamic Tanh

$$\text{DyT}(x) = \tanh(\alpha x)$$

- No batch dependence, no train/eval gap, no statistics.
- Observation: LayerNorm's input→output map is S-shaped, like a tanh.
- learns $\tanh(\alpha x)$. No batch/token statistics at all.
- Matches or beats normalized ViT / DiT / LLMs.

RMSNorm

Saves compute

$$\bar{a}_i = \frac{a_i}{\text{RMS}(\mathbf{a})} g_i, \quad \text{where } \text{RMS}(\mathbf{a}) = \sqrt{\frac{1}{n} \sum_{i=1}^n a_i^2}.$$

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16 / 22 acquired

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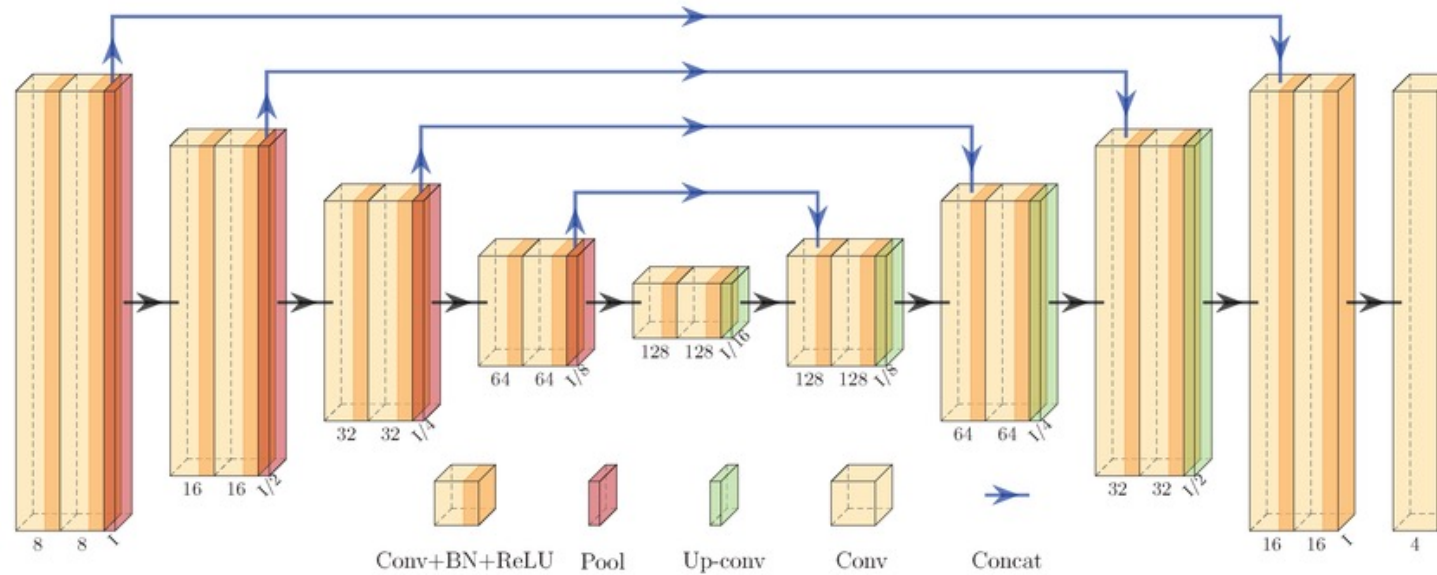
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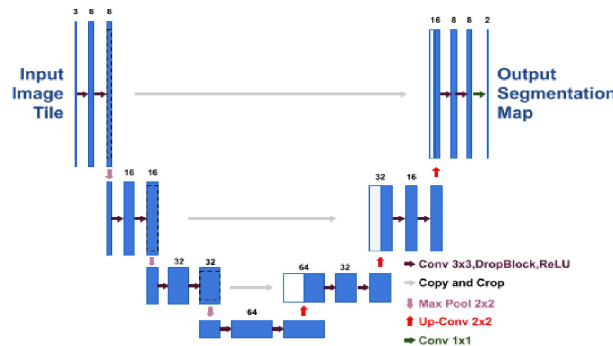
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U-Net



- Concat rather than add (significant?)
- For image/map outputs
- Common in diffusion models etc.
- Stupid name



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EXPLODING GRADS

Gradient clipping

cap $\|g\|$ by norm/value
tames spiky updates.

WHY?

High grads can create
irrecoverable modes

EVAL-TIME

Weight EMA

average weights over
training (Polyak).

WHY?

Optimization ends in a
distribution.

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19 / 22 acquired

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- GELU / SiLU

NORMALIZATION

- BatchNorm
- Layer / Group
- RMSNorm · DyT

REGULARIZATION

- L2
- dropout
- dropPath
- Weight decay
- augmentation
- label smooth · Mixup

OPTIMIZATION

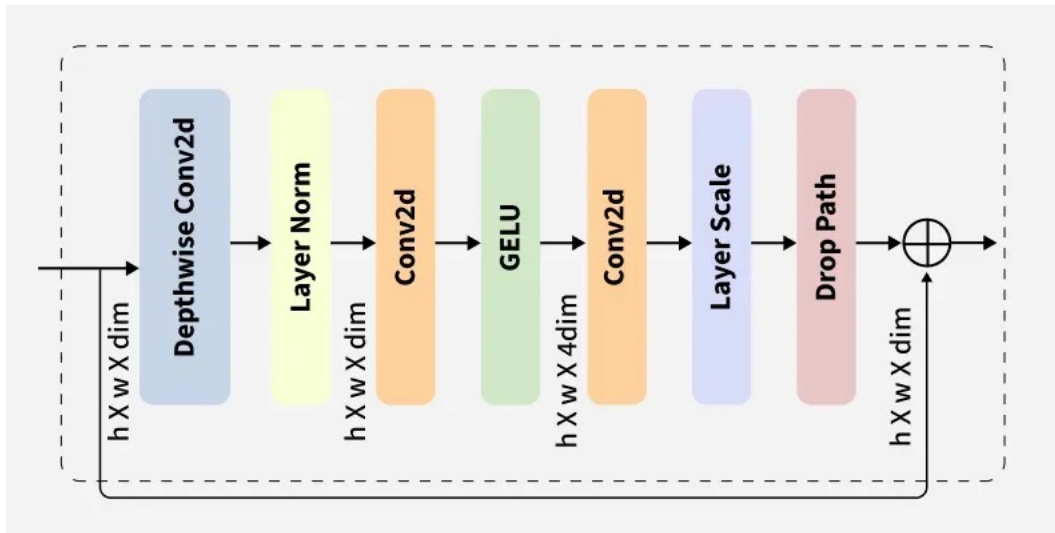
- SGD + momentum
- Nesterov
- Adam / AdamW
- Muon

GRADIENT TRICKS

- straight-through est.
- gradient clipping
- weight EMA

ConvNeXt 2022

- An improved recipe for CNN. Taking elements from ViT.
- Separate spatial mix and channel mix.
- One activation per block.
- LayerScale with tiny init vals.
- DropPath.
- GeLU, Layer-Norm



TOOLBOX

what we can build so far

20 / 22 acquired

ARCHITECTURE

- conv · pool · FC
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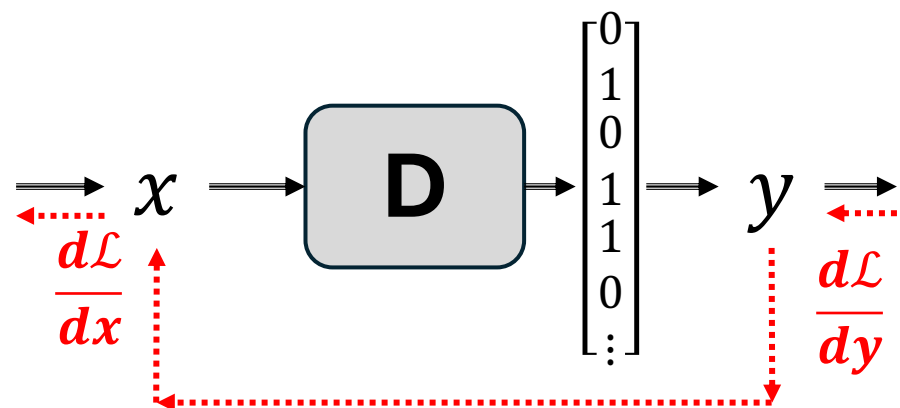
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Can we have discrete layers?



$$D = \begin{cases} \text{Round} \\ \text{Threshold} \\ \text{Nearest Neighbor} \\ \vdots \end{cases}$$

Autograd elegant trick:

$$y = (D(x) - x).detach() + x$$

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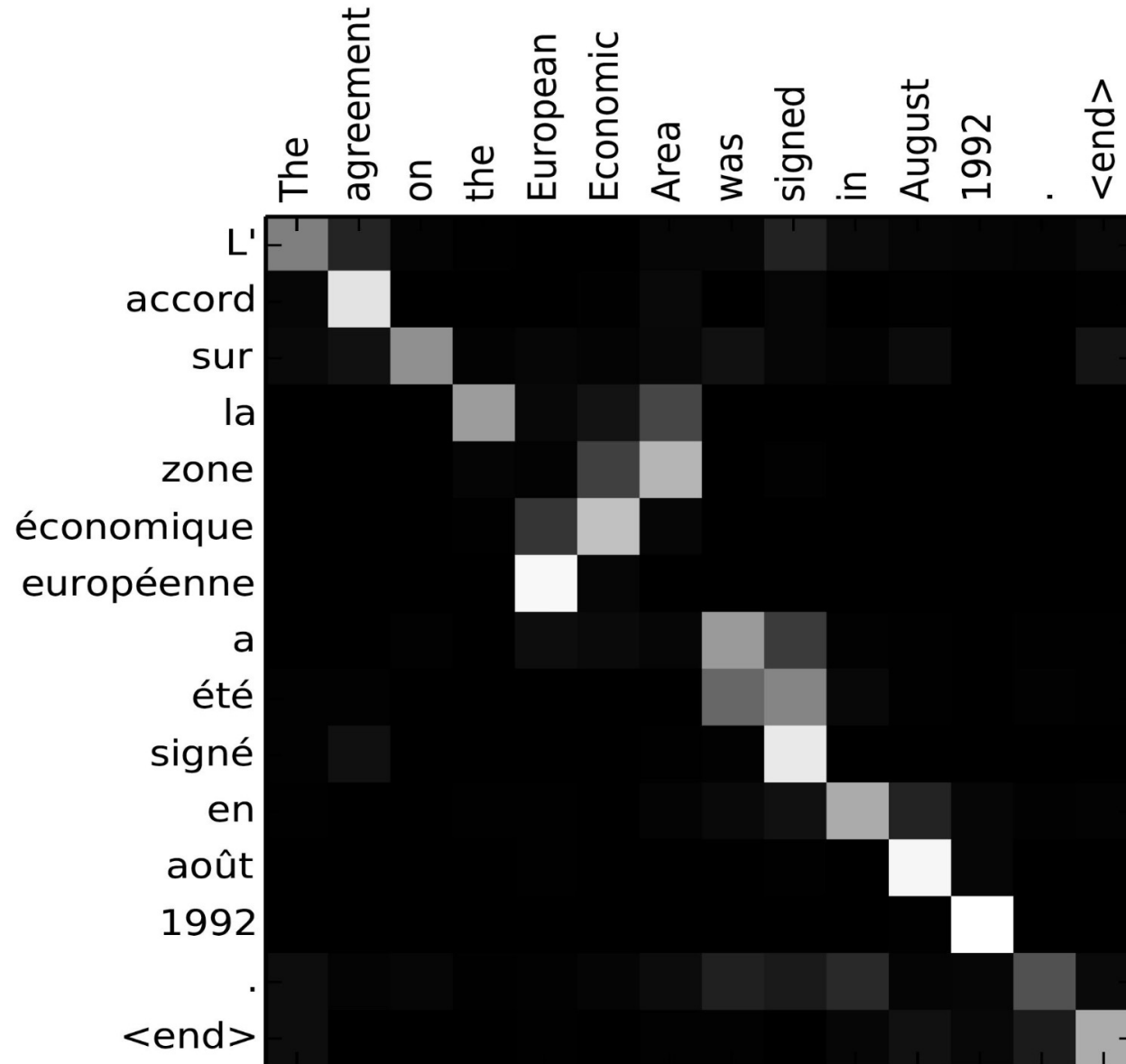
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Attention Mechanism



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(Self) Attention Mechanism

Inputs:

Input vectors: $\mathbf{X}$ (Shape: $N_x \times D_x$)

Key matrix: $\mathbf{W}_k$ (Shape: $D_x \times D_Q$)

Value matrix: $\mathbf{W}_v$ (Shape: $D_x \times D_V$)

Query matrix: $\mathbf{W}_Q$ (Shape: $D_x \times D_Q$)

Computation:

Query vectors: $\mathbf{Q} = \mathbf{XW}_Q$

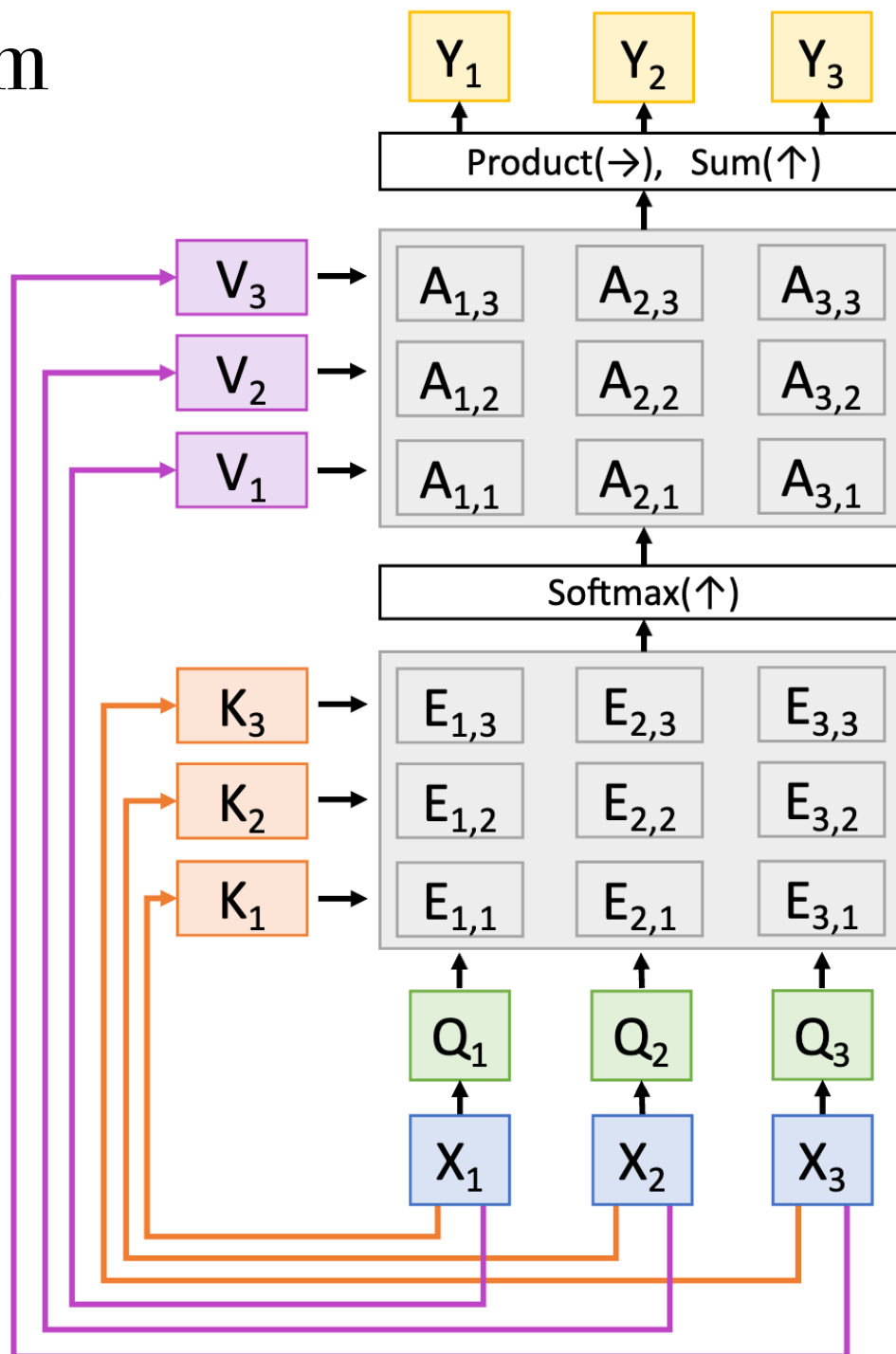
Key vectors: $\mathbf{K} = \mathbf{XW}_k$ (Shape: $N_x \times D_Q$)

Value Vectors: $\mathbf{V} = \mathbf{XW}_v$ (Shape: $N_x \times D_V$)

Similarities: $\mathbf{E} = \mathbf{QK}^T / \sqrt{D_Q}$ (Shape: $N_x \times N_x$) $E_{i,j} = (\mathbf{Q}_i \cdot \mathbf{K}_j) / \sqrt{D_Q}$

Attention weights: $\mathbf{A} = \text{softmax}(\mathbf{E}, \text{dim}=1)$ (Shape: $N_x \times N_x$)

Output vectors: $\mathbf{Y} = \mathbf{AV}$ (Shape: $N_x \times D_V$) $Y_i = \sum_j A_{i,j} \mathbf{V}_j$



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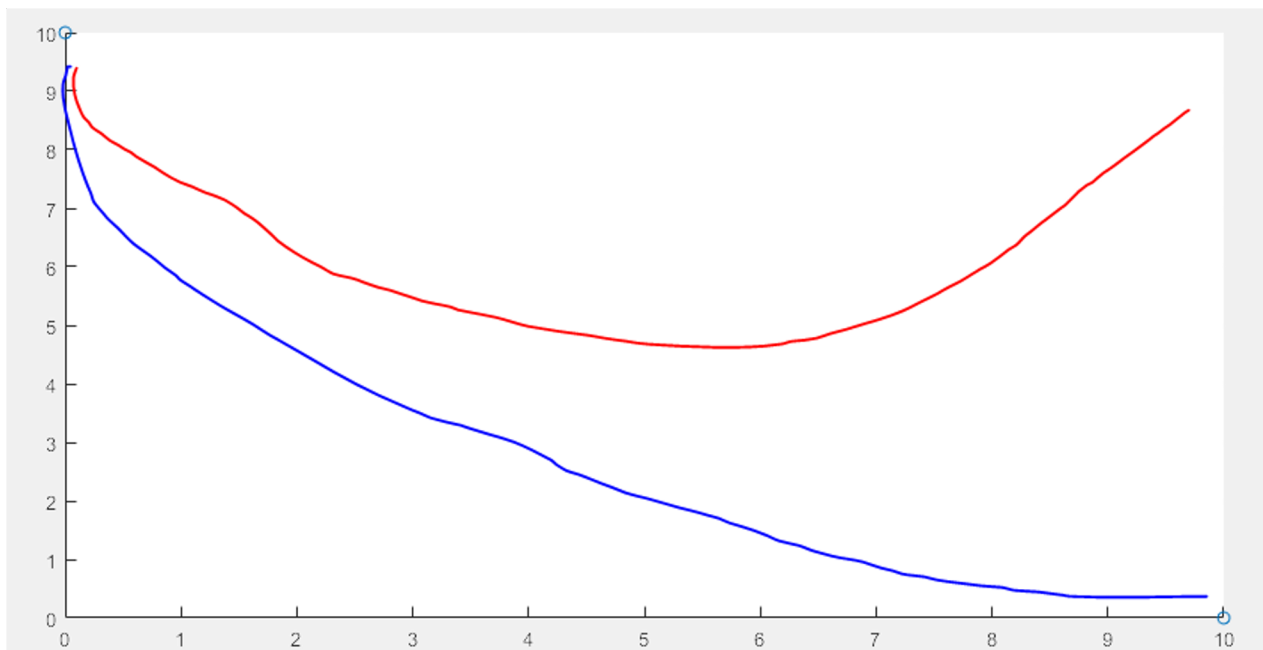
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What's happening and what should we do?



Overfitting:

- Regularize
- Reduce complexity
- Early stop (if happy with result)
- Smarter architecture
- Data augmentation

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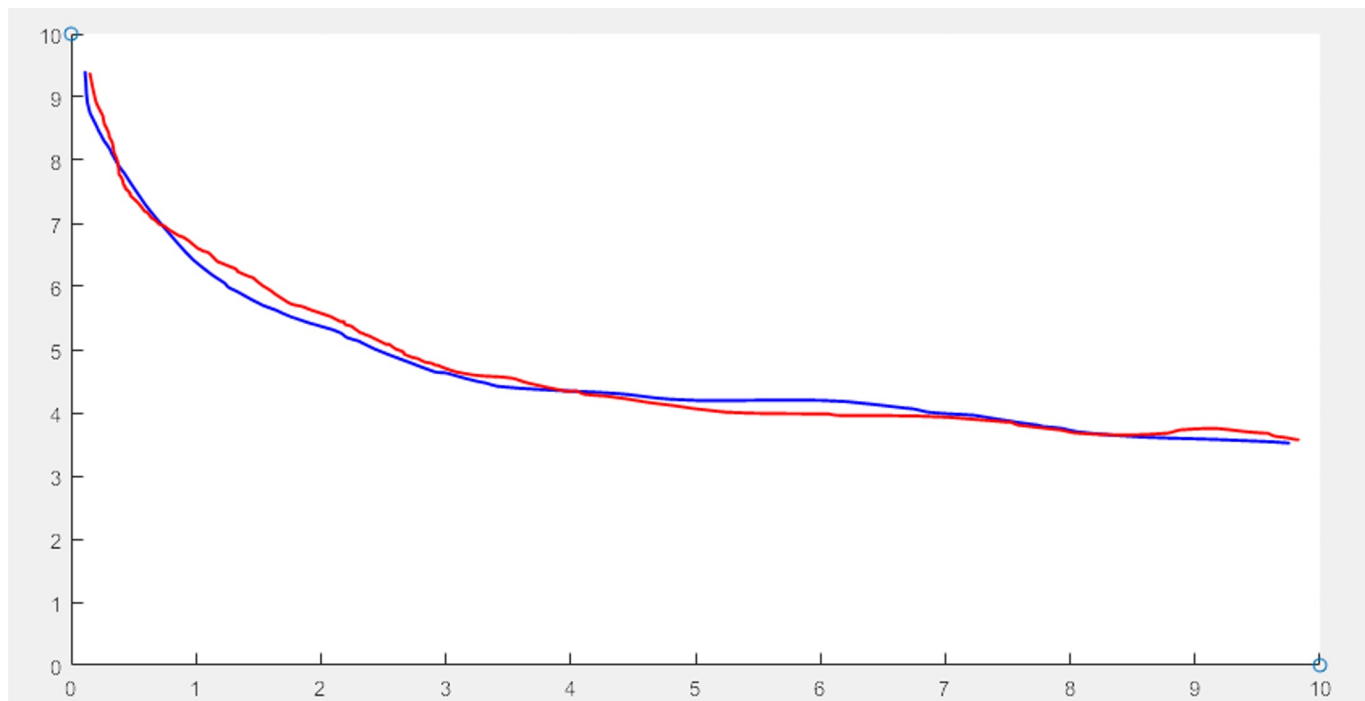
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What's happening and what should we do?



Underfitting:

- Stronger network

TOOLBOX

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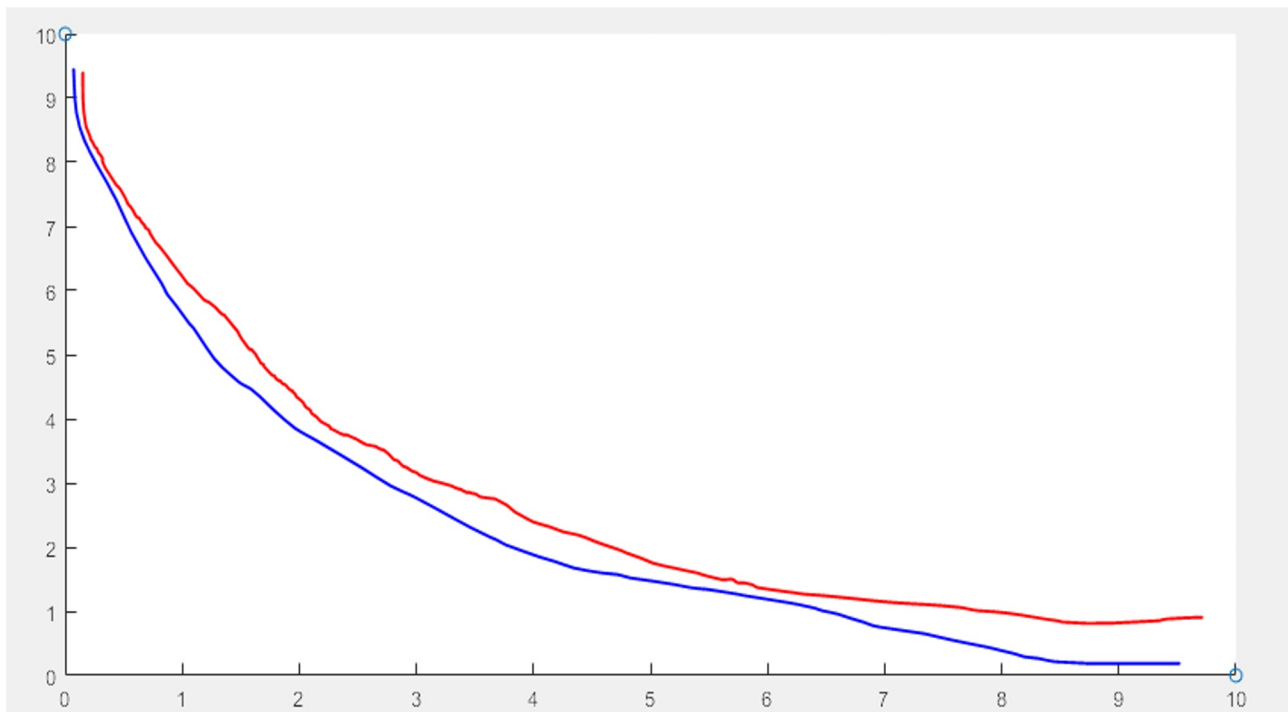
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What's happening and what should we do?



OK

- Typically you want some overfitting, because the sweet-spot of best test-error has a little fitting error and a little generalization error

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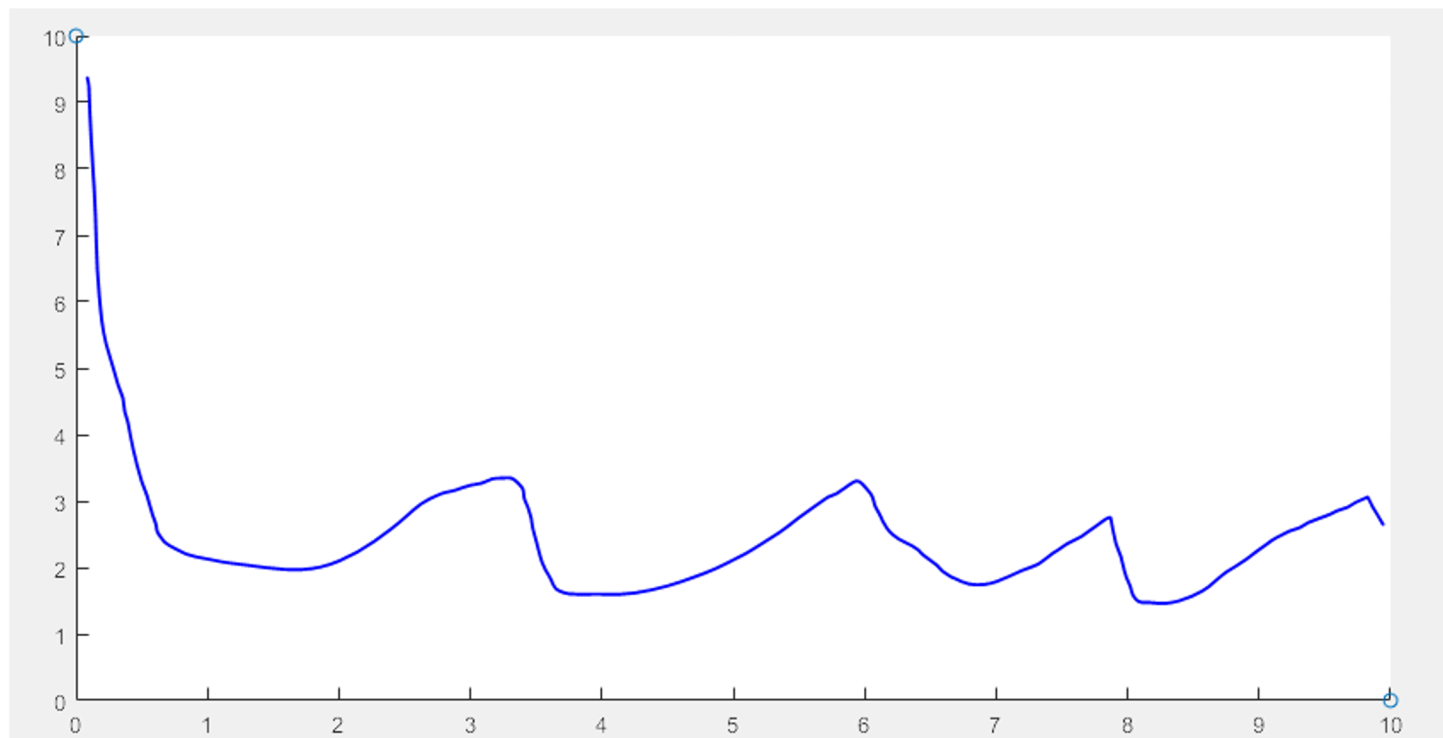
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What's happening and what should we do?



Train error too noisy:

- Bigger batch size
- Lower learning rate

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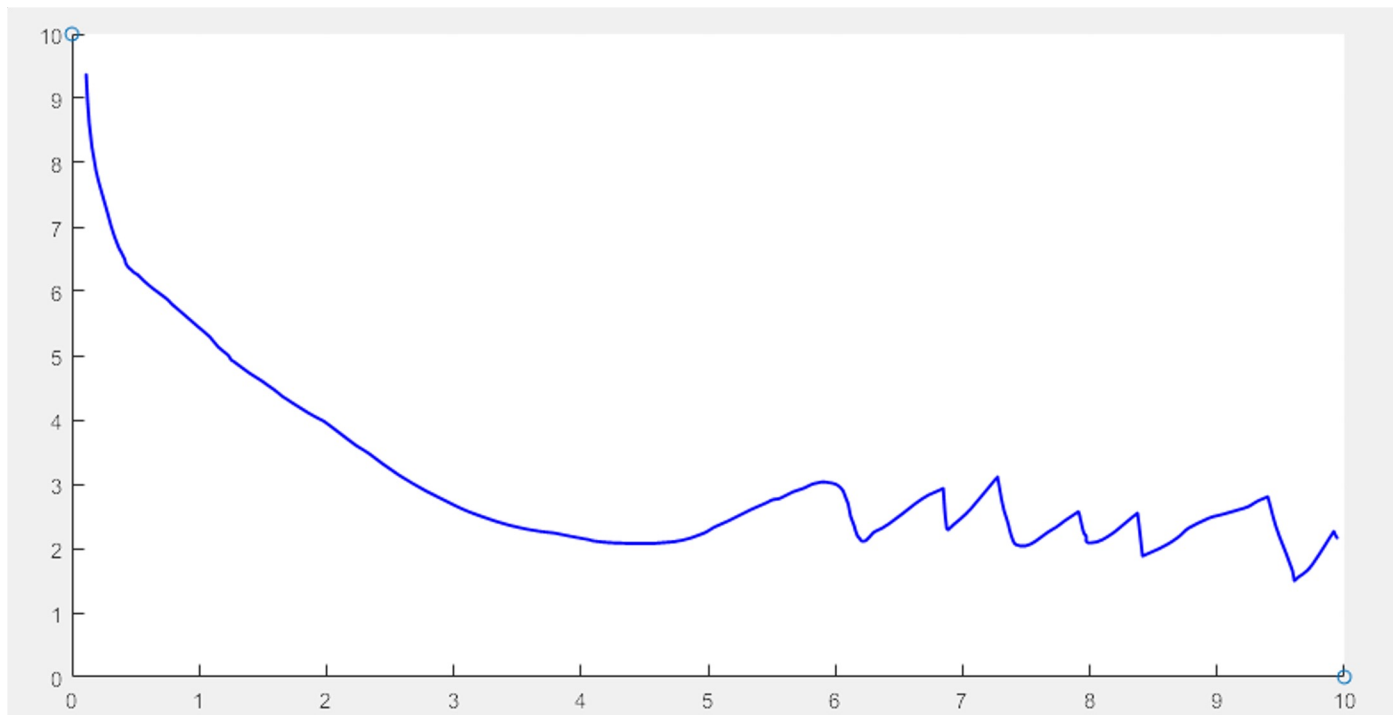
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What's happening and what should we do?



“Stuck in noise”

- Add a change in learning rate after some iterations or when stuck.

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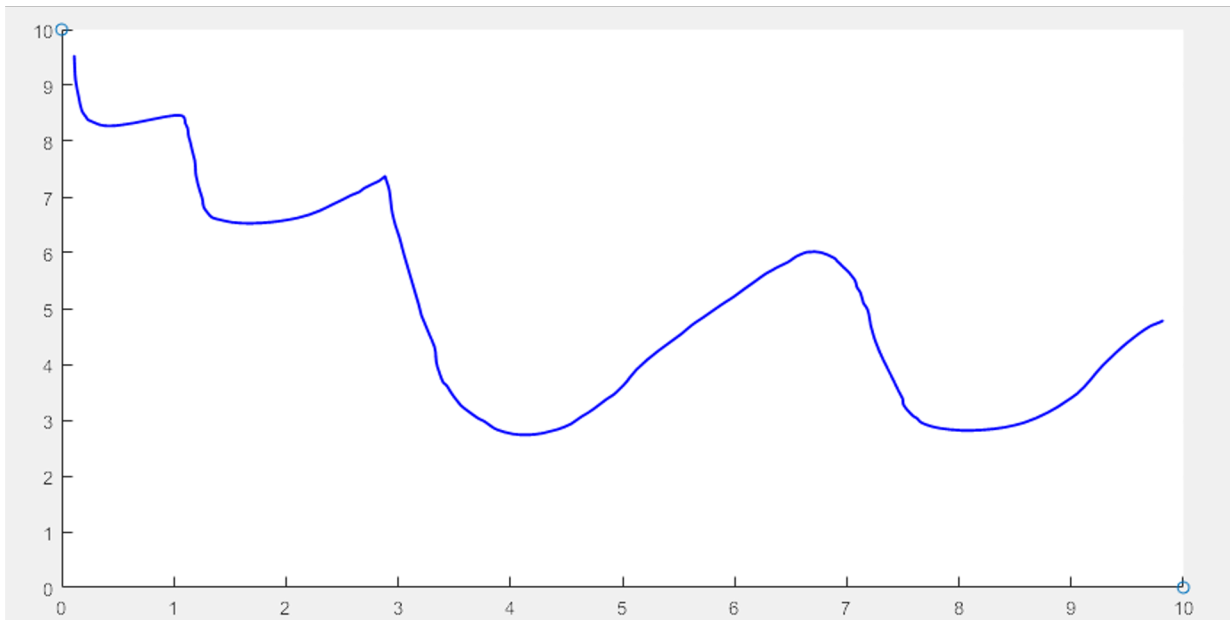
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What's happening and what should we do?



Optimization makes no sense:

- Debug
- Huge learning rate?
- Architecture failure?
- Initialization problem?
- Dead neurons? Use leaky ReLU?

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